Institutional Portfolios for Efficient Democratic Government

May 11, 2026

Abstract

This working paper asks what can be learned by combining the legislative, constitutional-review, and lobbying/capture simulator projects into one portfolio screen. The current revision rechecks 4 updated sibling papers and report artifacts, then imports 1,428 legislative campaign rows, 520 constitutional-review rows, and 31 lobbying/capture rows, producing 33,852 primary institutional portfolios. It does not treat every updated review artifact as interchangeable. Instead, it builds a review harmonization manifest, a companion review-source swap, and a conservative harmonized review source with 10 shared static metrics. The harness then runs a focused 24-period interbranch bridge model over 7 selected cases and a synthetic 36-period adaptive bridge over all 33,852 primary portfolios. The result is not a settled recommendation. Several candidate portfolios beat the stylized current-system baseline in the current synthetic screens, but no portfolio yet passes the adaptive recommendation gate. The balanced winner remains the leading static point-estimate and lower-bound candidate, while Adaptive Bridge v1 marks it do not recommend yet. The best use of the paper is therefore screening: narrow the design space, expose fragile winners, and specify the validation work needed before institutional recommendations are made.

1 Research Question

The question is not which single institution looks strongest in isolation. The question is which combinations preserve public lawmaking capacity while preventing predictable institutional failure modes:

- low-support or weak-mandate policy passage;
- rights-threatening laws or emergency actions;
- lobbying capture, dark influence, and reform evasion;
- constitutional instability and interbranch conflict;
- excessive administrative cost or procedural complexity.

The object of comparison is therefore the institutional portfolio: one legislative design, one review design, and one anti-capture design evaluated together.

The contribution is narrower than a constitutional blueprint. The paper shows how separately calibrated simulators can be combined without pretending they form one statistical model. It uses the combined screen to sort candidate families, identify fragile headline winners, and state what evidence would be required before stronger claims are justified.

2 Argument Map

The body keeps only claims that change how a skeptical reader should interpret the results. Other diagnostic material is moved to the appendix.

Claim role	Evidence used in the body	Permitted conclusion
Source scope	Imported rows, provenance, and sibling-paper crosswalk	The paper is a synthesis of three imported simulator families, with one updated constitutional-review companion used as a scope-control cross-check.
Static screen	Balanced score, uncertainty overlap, and the focused candidate comparison	Some portfolios outperform the stylized current-system baseline, but rank one is not uncertainty-resolved.
Feedback stress	Focused bridge v0 and stress exceptions	Candidate families must be judged by adaptation pressure, not only static efficiency.
Recommendation boundary	Adaptive Bridge v1 gate counts and validation agenda	Current winners are validation targets, not settled institutional recommendations.

The paper separates three claim types. Empirical-source claims are limited to the imported report artifacts and their recorded provenance. Synthetic findings are claims produced by cumulative scoring, regret, Pareto, robustness, and bridge layers. Speculative design recommendations are proposals for future model extensions, calibration work, or reform sequences that are not yet tested as portfolio rows or bridge mechanisms.

3 Evidence Base

The cumulative project imports three existing simulator families and cross-references one updated companion review project. The Congress Institutional Simulator contributes lawmaking capacity, public alignment, welfare, gridlock, legitimacy, concentrated-harm passage, lobbying pressure, and administrative-feasibility diagnostics. The Supreme Court Simulator Design project contributes rights protection, review stability, democratic responsiveness, emergency abuse control, independence/accountability balance, legitimacy, public confidence, constitutional conflict, and cost diagnostics. The Lobby Capture Simulator contributes capture control, representation, public-interest alignment, detection, sanctioning, net transparency, hidden influence, reform decay, and administrative-cost diagnostics.

The updated Constitutional Review Simulator now contributes a stronger comparative-review check on the paper's interpretation. It is not folded into the primary ranking as if it were just more rows from the same simulator, because it overlaps the imported review family while using different case structures, denominators, and additional compliance, implementation, defiance, emergency, and public-trust fields. A naive concatenation would overstate certainty. The current build therefore separates three objects: the primary imported Supreme Court Design source, a companion Constitutional Review source swap, and a conservative harmonized review source that includes only same-name, same-direction, defensibly comparable static review fields.

The reconciliation report finds 10 shared configured review metrics, 6 imported-only configured metrics, 2 companion-only configured metrics, and 16 companion fields that are better treated as adaptive inputs unless their denominators are harmonized. The manifest admits 10 static metrics

Source	Rows	Scenarios	Metrics	Imported report
Congress Institutional Simulator	1,428	42	18	`simulation-campaign-v21-paper.csv`
Supreme Court Simulator Design	520	26	18	`constitutional-review-campaign-v2.csv`
Lobby Capture Simulator	31	31	21	`lobby-capture-campaign.csv`

Table 1: Current source artifacts imported by the cumulative harness.

Project	Updated paper checked	Current report artifact	Cumulative use	Reading rule after refresh
Congress Institutional Simulator	Searching Legislative Design Space for Compromise and Productivity; `main.tex`; hash `2791c54c98b5`	`simulation-campaign-v21-paper.csv`; 1,428 rows; hash `1e36005c54d3`	Imported legislative source.	Treat the updated portfolio-hybrid legislature as a synthesized candidate; pairwise alternatives remain the cleaner non-default productivity and compromise comparator.
Supreme Court Simulator Design	Designing Constitutional Review Institutions by Simulation; `main.tex`; hash `eef2eeba51e1`	`constitutional-review-campaign-v2.csv`; 520 rows; hash `2278a169405c`	Imported constitutional-review source.	Keep review metrics pathway-specific and proxy-aware; the updated denominator audit does not permit pooled validation claims.
Constitutional Review Simulator	When Constitutional Review Improves Democratic Constitutionalism: A Comparative Simulation Framework; `main.tex`; hash `dc9291f0876e`	`constitutional-review-sensitivity-v1.csv`; 231 rows; hash `dc638127a5c1`	Referenced review cross-check; not imported into rankings yet.	Use the richer compliance, defiance, implementation, emergency, and public-trust fields to guide schema reconciliation before any row merge.
Lobby Capture Simulator	Strategic Channel Substitution and Regulatory Capture: A Simulation Model of Lobbying and Anti-Capture Reform; `main.tex`; hash `72e62312fe3c`	`lobby-capture-campaign.csv`; 31 rows; hash `8e1e313e166c`	Imported anti-capture source.	Read low observed capture through total distortion, hidden influence, venue-shifting, and validation partial/miss diagnostics.

Table 2: Crosswalk from updated sibling simulator papers and reports to the cumulative synthesis. The Constitutional Review Simulator is deliberately treated as a review cross-check until its richer schema is reconciled with the imported Supreme Court Simulator Design campaign.

Review reconciliation category	Count
shared configured metric	10
imported-only configured metric	6
companion-only configured metric	2
missing from both review sources	0
companion adaptive candidate	16

Table 3: Construct-level reconciliation between the imported Supreme Court Simulator Design review source and the companion Constitutional Review Simulator. Companion adaptive candidates are not added to the static ranking until denominator and construct ownership are resolved.

Harmonization action	Manifest rows
included	20
excluded	8
not present	8
adaptive candidate	16

Table 4: Review harmonization manifest. Included static fields are: constitutionalConflict, democraticResponsiveness, directionalScore, independenceAccountabilityBalance, legalStability, legitimacy, partisanAlignment, reversalRate, rightsProtection, shadowDocketAbuse. Source-specific fields remain available in source-swap sensitivity screens or adaptive calibration work, but are excluded from the harmonized static source.

into the harmonized source and excludes source-specific composites, public-confidence/cost fields, and compliance fields from the harmonized static score. The integration problem is therefore not just missing data. It is construct ownership. These source simulators were calibrated separately, so the cumulative score cannot be read as a single unified statistical estimate. It is a structured comparison across separately generated artifacts. The source-paper crosswalk records which updated artifacts are imported, cross-checked, and harmonized. For that reason, the paper treats scores as screening evidence and then checks whether the same candidates survive uncertainty, value-profile, bridge, stress, adaptive, and review-source filters.

4 Cumulative Method

The cumulative harness groups each source report by scenario, computes component scores from available columns, and builds every legislature-review-anti-capture combination. It then compares portfolios on delivery, public alignment, rights safeguard, capture resistance, legitimacy, efficiency, complexity, and resilience floor. The resilience floor is deliberately conservative: a portfolio that performs well in two subsystems but collapses in capture resistance, rights protection, or complexity should not outrank a more balanced package without an explicit normative reason.

The harness reports 6 scoring profiles plus complementary Pareto, robustness, bridge, and bridge-sensitivity views. This matters because a single blended score can hide the strongest design lesson: different constitutional values reward different institutional packages, and some high-throughput packages look worse once rights, capture, legitimacy, and implementation constraints are treated as floors.

5 Which Designs Beat the Current-System Baseline?

The phrase "beat the current system" has a narrow meaning here. The current-system-ish row is a stylized benchmark assembled from the imported simulator families. It is not an empirical claim about the live U.S. constitutional order. A candidate beats that row only when it performs better in the current synthetic screens.

The central result is that the stylized baseline is not competitive in the combined screen, while the leading alternatives separate into different roles. The balanced static winner is the best headline screen result. The robustness challenger is the stronger validation target once profile durability and stress regret matter. The adaptive gray-zone leader is the best all-portfolio adaptive screen result. None of those labels is a recommendation, because the adaptive gate produces 0 provisional shortlist rows and 4,833 calibration gray-zone rows.

Candidate	Why it is in the paper	Static screen	Feedback/adaptive screen	Paper status
Current-system-ish baseline	Stylized comparator, not an empirical claim about the live U.S. system.	rank 33,486; score 0.500; floor 0.419	bridge 7/7 (0.474); adaptive rank 14,704	Does not beat the leading synthetic portfolios; kept to show the comparison floor.
Balanced static winner	Best point-estimate and lower-bound static screen.	rank 1; score 0.656; max regret 0.107	bridge 1/7 (0.723); adaptive rank 3,320	Beats the stylized baseline in static and focused-bridge screens, but gate is do not recommend yet.
Robustness challenger	Best cross-profile robustness row and strongest stress-regret challenger.	balanced rank 267; robustness rank 1; floor 0.635	bridge 6/7 (0.709); adaptive rank 7	Stronger validation target than the static winner under adaptive screening; gate is calibration gray zone, not recommendation.
Adaptive gray-zone leader	Highest all-portfolio adaptive-survival score.	static rank 125; score 0.640; max regret 0.083	bridge 4/7 (0.713); adaptive rank 1	Best current validation target, not a recommendation; gate is calibration gray zone, not recommendation.

Table 5: Reader-facing synthesis of the stylized current-system baseline and the main challenger portfolios. “Beats” means stronger in the current synthetic screens, not empirically proven superior.

6 Static Finding

The balanced-score winner is:

> Expanded portfolio hybrid legislature + No emergency relief without merits review + Full anti-capture bundle.

Portfolio component	Selected design
Legislature	Expanded portfolio hybrid legislature
Review	No emergency relief without merits review
Anti-capture	Full anti-capture bundle

Diagnostic	Value
Balanced score	0.656
Policy delivery	0.745
Public alignment	0.534
Rights safeguard	0.724
Capture resistance	0.648
Legitimacy	0.613
Complexity score	0.554
Resilience floor	0.554

Table 6: Balanced-score winner and diagnostics.

Its balanced score is 0.656. The diagnostic pattern is strong delivery (0.745), strong rights safeguard (0.724), moderate capture resistance (0.648), moderate legitimacy (0.613), moderate complexity score (0.554), and lower public alignment (0.534). It is a plausible headline candidate under the point-estimate score because it does not rely on one institutional mechanism alone. It combines a high-capacity legislative portfolio, a review rule that sharply restrains emergency relief, and a lobbying environment that allows access under stronger anti-capture controls.

The weakness is public alignment. The balanced winner should not be framed as a fully representative ideal. It is better described as a high-delivery, high-rights-screen portfolio with adequate capture control and manageable complexity. Its synthetic uncertainty interval is 0.563–0.748 around the current score, with half-width 0.092. Its lower-bound rank is 1, but 33,852 portfolios overlap its interval. Its minimax-regret rank is 5,956 and its maximum profile regret is 0.107. Those diagnostics keep it in the lead under current scoring, but they prevent treating the rank-one row as uncertainty-resolved.

The uncertainty bands below are not empirical confidence intervals. They are conservative model-sensitivity bands. They widen when a portfolio combines separately calibrated source artifacts, shows component imbalance, has cross-profile instability, or depends on thin metric coverage.

Uncertainty diagnostic	Value
Balanced interval	0.563–0.748
Rows overlapping balanced interval	33,852
Rows entirely below balanced interval	0
Rows entirely above balanced interval	0
Balanced lower-bound rank	1
Lower-bound leader	Expanded portfolio hybrid legislature + No emergency relief without merits review + Full anti-capture bundle (balanced rank 1)

Table 7: Interval-resolution diagnostics for all 33,852 portfolios. Overlap counts show whether point-estimate ranks are separated by the synthetic uncertainty bands.

The interval-resolution diagnostic is severe: 33,852 generated portfolios overlap the balanced winner's interval. The lower-bound ordering still places Expanded portfolio hybrid legislature + No emergency relief without merits review + Full anti-capture bundle first, but the project should describe that as a robustness clue, not as statistical separation from the field.

The most robust cross-profile portfolio is:

> Portfolio hybrid legislature + No emergency relief without merits review + Full anti-capture bundle.

It ranks only 267 under the balanced score, but it has the best cross-profile durability: average profile rank 1538.833, worst profile rank 4,727, and top-25 placement in 3 of the 6 scoring profiles. Substantively, that points toward a design hypothesis: structured alternative selection may be more robust than a broad expanded hybrid when the evaluator cares about legitimacy, rights, and anti-capture goals as much as headline delivery. Adaptive Bridge v1 also ranks this robustness candidate much higher than the static balanced winner: adaptive rank 7 versus 3,320.

The balanced winner itself remains provisional because it has weak public-alignment diagnostics and falls far under at least one value profile. The efficiency and minimax-regret leaders are more fragile: they reduce regret or increase throughput, but their default-passage structure makes them vulnerable to weak-floor, rights, legitimacy, or capture concerns. These rows are useful for learning what the scoring model rewards; they are not ready as constitutional recommendations.

7 Feedback and Stress Findings

The bridge model moves the project beyond static combination, but only for a focused candidate set. It selects 7 cases from 33,852 static portfolios: balanced winner, minimax-regret winner, robustness winner, rights/capture winner, legitimacy-first winner, efficiency caution case, and current-system-ish baseline. The remaining 33,845 static portfolios are not bridge-simulated in this paper workflow.

Within that focused set, the baseline bridge winner is Balanced winner:

> Expanded portfolio hybrid legislature + No emergency relief without merits review + Full anti-capture bundle.

Rank	Case	Score	Avg. delivery	Final quality	Final alignment
1	Balanced winner	0.723	0.601	0.350	0.516
2	Efficiency caution case	0.719	0.627	0.345	0.490
3	Minimax-regret winner	0.715	0.595	0.337	0.505
4	Rights/capture winner	0.713	0.528	0.305	0.507
5	Legitimacy-first winner	0.713	0.528	0.305	0.507
6	Robustness winner	0.709	0.516	0.298	0.507
7	Current-system-ish baseline	0.474	0.297	0.000	0.195

Table 8: Baseline interbranch bridge ranking.

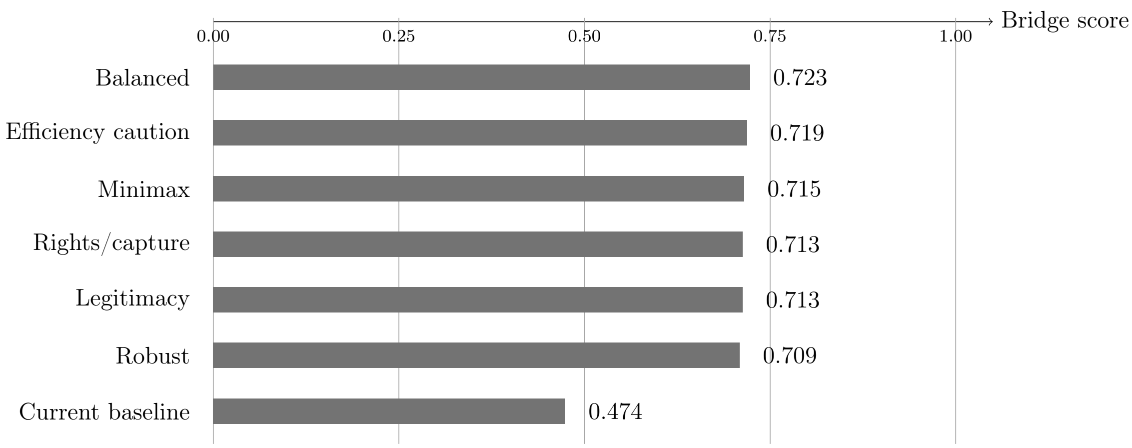

Figure 1: Baseline bridge scores for the focused portfolio cases.

This matters because the earlier robustness result could have suggested replacing the balanced winner with the pairwise-alternatives family. Focused bridge v0 does not support that as a clean replacement under baseline assumptions. Instead, it shows a tight top cluster. The balanced winner keeps more average delivery (0.601) and the highest final policy quality in the baseline bridge run (0.350), while the minimax, legitimacy, robustness, and rights/capture cases remain close enough that the bridge should be read as a comparison-set filter rather than as a global portfolio ranking.

The current-system-ish baseline performs worst in baseline bridge v0. It has low average delivery (0.297), final policy quality of 0.000, and the report flags weak starting floor, low policy quality, public-alignment erosion, and delivery bottleneck. This result should be treated as a stylized baseline comparison, not as an empirical claim about any real constitution.

Bridge v0 reruns the same 7 cases under 8 configured stress profiles. By the configured wins/top-two rule, the balanced winner ranks first: it wins 5 of 8 stress profiles, places top-two in 5, and has average stress rank 2.250. The stability conclusion still depends on the rule. Robustness winner has the best average bridge score and Robustness winner has the lowest maximum stress regret, so wins/top-two should not be described as the only defensible stability criterion.

The major exceptions are high capture pressure and emergency abuse. Under high capture pressure, Robustness winner takes first:

> Portfolio hybrid legislature + No emergency relief without merits review + Full anti-capture bundle.

The balanced winner falls to rank 5 under high-capture stress and is flagged for low policy quality, public-alignment erosion, capture drift, and legitimacy erosion. That is the most important caveat in the current configuration. The balanced winner is strong when capture controls remain effective

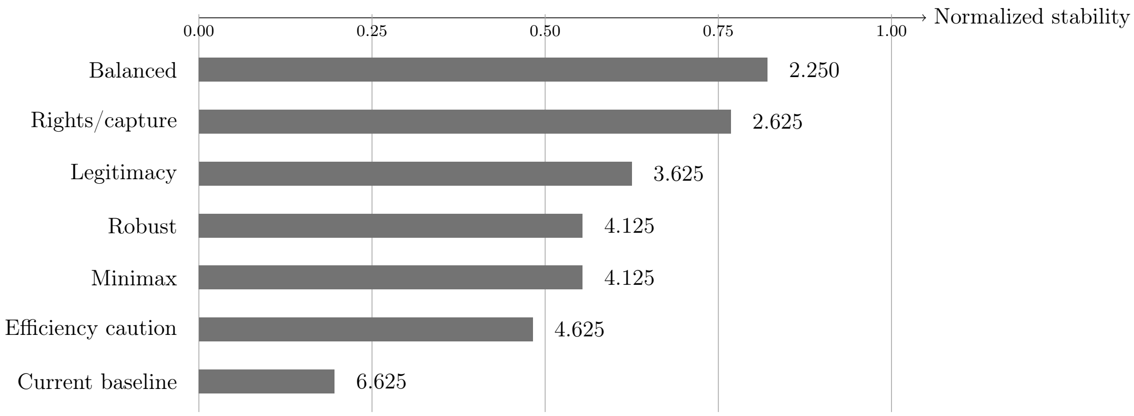

Figure 2: Average-rank stress stability across the focused bridge cases. Bar labels show average stress rank; lower rank is better.

enough, but it can degrade sharply when organized interests adapt faster than the anti-capture layer can suppress.

Under emergency-abuse stress, the Robustness winner takes first and the balanced winner falls to rank 4. The failure mode is different from high capture: the balanced winner is not primarily losing because capture overwhelms it, but because emergency-driven rights pressure and correction backlog exceed the review/correction channel in the bridge assumptions. Under federalism and agency-capacity stress, the balanced winner remains first but carries a weak-starting-floor flag, so that stress should be treated as a modeling priority rather than a solved condition.

Rank	Case	Score	Final quality	Final capture	Final legitimacy	Failure flags
1	Robustness winner	0.479	0.003	0.573	0.420	low policy quality; public alignment erosion; capture drift; legitimacy erosion
2	Rights/capture winner	0.446	0.000	0.661	0.389	low policy quality; public alignment erosion; capture drift; legitimacy erosion
3	Legitimacy-first winner	0.446	0.000	0.661	0.389	low policy quality; public alignment erosion; capture drift; legitimacy erosion
4	Current-system-ish baseline	0.304	0.000	1.000	0.129	weak starting floor; low policy quality; public alignment erosion; capture drift; legitimacy erosion; delivery bottleneck
5	Balanced winner	0.189	0.000	1.000	0.095	weak starting floor; low policy quality; public alignment erosion; capture drift; uncorrected rights backlog; legitimacy erosion
6	Minimax-regret winner	0.180	0.000	1.000	0.059	low policy quality; public alignment erosion; capture drift; uncorrected rights backlog; legitimacy erosion
7	Efficiency caution case	0.165	0.000	1.000	0.000	low policy quality; public alignment erosion; capture drift; uncorrected rights backlog; legitimacy erosion

Table 9: High-capture stress diagnostics.

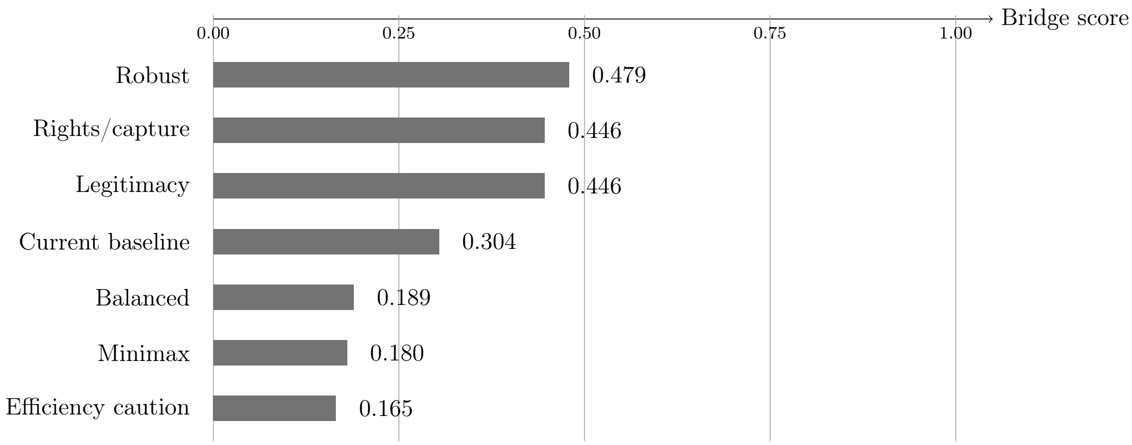

Figure 3: High-capture bridge scores across the focused bridge cases.

8 Adaptive Survival Screen

Adaptive Bridge v1 is a broader but more speculative screen. It runs 270,816 portfolio-stress simulations: every one of the 33,852 static portfolios under the 8 stress profiles, for 36 periods each. It keeps the v0 stress labels, then adds synthetic transition load, sequencing readiness, party adaptation, court adaptation, lobby venue shifting, emergency safeguards, federalism resistance, delivery bottlenecks, recovery capacity, voter feedback, and citizen agenda capacity.

This is not empirical validation. It is a skeptical institutional-survival screen. The Deep Research reports justify directional priors: narrow lobbying constraints invite venue shifting; emergency review matters only if it is fast, reason-giving, and anti-evasion oriented; high-complexity reforms need sequencing and administrative capacity; federalism and delivery bottlenecks are distinct failure modes; citizen agenda channels work best when deliberative and authenticated; and trust erosion is easier than trust recovery. The screen uses those priors to ask whether a portfolio still looks plausible after parties, courts, lobbyists, agencies, voters, and public agenda channels adapt.

The current adaptive leader is:

> Unicameral majority + pairwise alternatives + No emergency relief without merits review + Full anti-capture bundle.

Its average adaptive score is 0.705, worst stress rank is 34, maximum adaptive stress regret is 0.021, and the gate labels it calibration gray zone, not recommendation. The current run has 0 provisional shortlist rows, 4,833 calibration gray-zone rows, 0 review-priority rows, and 29,019 rows marked do not recommend yet.

The important negative finding is that the static balanced winner does not survive this harsher adaptive screen. It falls to adaptive rank 3,320, has average adaptive score 0.648, worst stress rank 14,897, and maximum adaptive stress regret 0.172. Its gate is do not recommend yet. That does not refute the static score; it means the static balanced winner depends on assumptions that become fragile when high capture, emergency abuse, low trust, noncompliance, and implementation overload are modeled as adaptive processes.

The robustness candidate is stronger under v1 but still not a recommendation. It ranks 7 with average adaptive score 0.700 and worst stress rank 117; its gate is calibration gray zone, not recommendation. The v1 result therefore shifts the paper's comparative focus from one balanced point-estimate winner to a family comparison: pairwise-alternatives and pairwise-tournament legislatures with emergency restraint and full anti-capture, audit, or voucher supports, plus portfolio-hybrid

Adaptive gate	Count	Highest-ranked example
provisional shortlist	0	none
calibration gray zone, not recommendation	4,833	Unicameral majority + pairwise alternatives + No emergency relief without merits review + Full anti-capture bundle (rank 1)
review priority, not recommendation	0	none
do not recommend yet	29,019	Citizen initiative and referendum + Pre-enactment constitutional council + Full anti-capture bundle (rank 135)

Table 10: Recommendation gate counts from Adaptive Bridge v1. Rows outside the provisional shortlist remain research targets rather than institutional recommendations.

challengers retained by the static robustness diagnostics. That comparison is a research agenda, not a settled recommendation, because pairwise, portfolio-hybrid, and budgeted-lobbying families still need direct family-specific validation.

9 Design Thesis

Within this stylized evidence base, the most promising configured direction is not a minimalist majoritarian system or a maximal-review system. The emerging design direction is a moderated institutional portfolio:

- a legislature with meaningful agenda access, alternative comparison, public-interest filtering, harm-sensitive safeguards, and active-law correction;
- a review system with enough independence to protect rights, strong emergency-restraint rules, and enough transparency to reduce legitimacy collapse;
- anti-capture rules that combine disclosure, public participation, enforcement, and evasion-aware monitoring without making ordinary advocacy impossible.

10 Validation Agenda

The current rows test institutional portfolios assembled from the imported simulators. Adaptive Bridge v1 now adds first-pass synthetic versions of transition load, implementation capacity, public feedback, citizen agenda channels, and adaptive political behavior. Those additions are useful, but they are still not deep enough to settle institutional recommendations. The next systems to model are speculative until they are implemented in source simulators, calibrated against evidence, and stress-tested as report artifacts.

A credible future version should explain why a candidate survives adaptation by parties, courts, lobbyists, agencies, and voters. Parties may alter agenda strategy. Courts may change docket behavior. Lobbyists may shift venues and messengers. Agencies may fail to implement complex reforms. Voters may punish or reward outputs in ways that change compliance and legitimacy. Adaptive Bridge v1 now exposes its mechanism-family coefficients in configuration and reports, but those coefficients are still evidence-anchored priors rather than fitted causal estimates.

The highest-priority source refactor remains the review layer, but the first harmonization pass is now explicit. The updated Supreme Court Simulator Design paper warns against pooling pathway denominators, while the updated Constitutional Review Simulator exposes richer compliance

Direction	Why add it	Claim status
Phased reform sequences	Model transition paths rather than assuming all institutional changes arrive at once.	Speculative extension
Constitutional transition costs	Represent amendment difficulty, legitimacy shocks, retraining costs, litigation delay, and path-dependence.	Bridge v1 first-pass synthetic; needs calibration
Federalism and agency capacity	Add subnational veto points, administrative staffing, enforcement capacity, procurement pressure, and intergovernmental conflict.	Bridge v0/v1 stress layer; needs source simulator
Public implementation feedback	Let public trust, compliance, and perceived policy quality feed back into future agenda access and reform durability.	Bridge v1 first-pass synthetic; needs calibration
Citizen agenda setting	Test ballot access, deliberative citizen panels, public agenda petitions, and participatory budget channels as proposal-entry systems.	Bridge v1 first-pass synthetic; needs mechanism rows

Table 11: Systems and mechanisms that should be modeled next before stronger institutional recommendations are made.

and implementation outputs. The current harmonized source uses only shared static fields; future builds should either justify replacing source-specific metrics, move compliance and implementation variables into adaptive inputs, or keep the primary ranking source unchanged.

The current build reruns three review variants: the primary Supreme Court Design source, the companion Constitutional Review source, and the harmonized review source. The companion run produces 27,342 portfolios; only 2 of its top 25 portfolios overlap the current imported-review top 25, and the current primary winner's rank in the companion run is not comparable. The harmonized run produces 61,194 portfolios; only 2 of its top 25 portfolios overlap the current imported-review top 25, its top-100 overlap with the current run is 6, and the current primary winner ranks 265 in that run. The harmonized winner is Expanded portfolio hybrid legislature + Constitutional Review Simulator: Mandatory legislative response cycles + Full anti-capture bundle. This is evidence that review-source choice materially affects the shortlist.

Run	Status	Review source	Scen.	Ports.	Score	Top-25 overlap	Primary winner rank	Run winner primary rank	Notes
current-imported-review	completed	Supreme Court Simulator Design	26	33852	0.656	25	1	1	Current ranking source.
companion-review-first-pass	completed	Constitutional Review Simulator	21	27342	0.673	2	not comparable	not comparable	First-pass source swap; missing configured fields are skipped, not imputed.
harmonized-review-source	completed	Harmonized review source	47	61194	0.671	2	265	not comparable	Conservative harmonized source using only shared same-direction static review fields from both review projects.

Table 12: Review-source static sensitivity. The companion run swaps in the Constitutional Review Simulator sensitivity campaign; the harmonized run uses only shared same-direction static review fields from both review projects. Overlap and rank columns use canonical portfolio keys with harmonized source prefixes stripped.

Left run	Right run	Top-25 overlap	Top-100 overlap	Shared canonical portfolios
current	companion	2	6	2,604
current	harmonized	2	6	33,852
companion	harmonized	19	86	27,342

Table 13: Pairwise overlap among review-source variants. Counts use canonical portfolio keys, so harmonized rows from prefixed source scenarios can be compared to the corresponding source-run scenario when one exists.

The goal is not to eliminate politics. The goal is to make strategic behavior visible, channel

Current	Companion	Harmonized	Comp.-cur.	Harm.-cur.	Portfolio
–	1	2	–	–	Expanded portfolio hybrid legislature + Suspended declarations of invalidity + Full anti-capture bundle
1	–	265	–	264	Expanded portfolio hybrid legislature + No emergency relief without merits review + Full anti-capture bundle
–	2	1	–	–	Expanded portfolio hybrid legislature + Mandatory legislative response cycles + Full anti-capture bundle
2	–	271	–	269	Expanded portfolio hybrid legislature + Emergency integrity package + Full anti-capture bundle
–	3	7	–	–	Expanded portfolio hybrid legislature + Weak-form review with legislative reply + Full anti-capture bundle
3	–	306	–	303	Expanded portfolio hybrid legislature + Automatic merits follow-up for emergency relief + Full anti-capture bundle
–	4	3	–	–	Expanded portfolio hybrid legislature + Rights-impact statements before review + Full anti-capture bundle
–	9	4	–	–	Expanded portfolio hybrid legislature + Hybrid court balancing independence and accountability + Full anti-capture bundle
4	–	336	–	332	Expanded portfolio hybrid legislature + Constitutional council with concrete-review backstop + Full anti-capture bundle
5	13	11	8	6	Expanded portfolio hybrid legislature + Pre-enactment constitutional council + Full anti-capture bundle
–	5	5	–	–	Expanded portfolio hybrid legislature + Abstract review tribunal + Full anti-capture bundle
–	6	9	–	–	Expanded portfolio hybrid legislature + Pre-enactment review before laws take effect + Full anti-capture bundle

Table 14: Rank shifts for the best-observed portfolios across review-source variants. Blank cells mean the canonical review scenario is not available in that variant.

it through contestable procedures, and prevent any one actor class from converting temporary advantage into durable institutional control.

A stronger future version needs external calibration before it can move from screening to recommendation. The minimum validation program is:

- historical data on proposal access, amendment selection, enactment, correction, and repeal under different legislative rules;
- emergency docket timing, reason-giving, merits follow-up, rights-impact, and compliance data for review systems;
- lobbying venue-shifting, enforcement, disclosure, sanctions, public-finance, and dark-money adaptation data;
- federalism, agency-capacity, implementation-delay, and public-feedback data that can calibrate transition and overload parameters;
- validation cases where parties, courts, lobbyists, agencies, and voters visibly adapted to a reform;
- external coefficient ranges for the configured Adaptive Bridge v1 mechanism families rather than only directionally justified priors.

11 Limitations

This paper does not claim that the current scores are empirically validated estimates of real-world constitutional performance. The sibling simulators are stylized comparative tools. The cumulative harness is a synthesis layer over their outputs.

The updated sibling papers do not all enter the ranking in the same way. Congress and Lobby Capture updates change imported CSV inputs directly. Supreme Court Simulator Design remains the imported review source. The updated Constitutional Review Simulator is referenced but not ranked because treating two differently calibrated review projects as interchangeable would double-count review evidence and mask denominator differences.

Bridge v0 is also stylized. Its feedback equations are transparent and auditable through `config/interbranch-bridge-v0.json`, but they are not calibrated causal estimates. It uses portfolio diagnostics as inputs rather than rerunning the sibling simulators period by period.

Adaptive Bridge v1 is broader than v0 but not more empirically authoritative. Its equations and mechanism-family coefficients are transparent through `config/adaptive-bridge-v1.json` and its reports, but its party, court, lobby, agency, voter, transition, and citizen-agenda mechanisms remain evidence-anchored priors rather than fitted estimates. It should be used to demote fragile winners and identify research priorities, not to announce a settled constitutional design.

The new uncertainty and regret diagnostics are guardrails, not solutions. The uncertainty bands are synthetic and should not be quoted as statistical confidence intervals. Because 33,852 portfolio intervals overlap the balanced winner's interval, point-estimate rank should be treated as a screening device rather than a settled ranking. The minimax-regret table depends on the currently configured value profiles; a different profile set could change the regret ordering. The watchlist identifies fragility inside the current model, but it cannot discover mechanisms the model has not implemented.

The adjacent `../Civ Pro` project is not imported yet because it is currently a civil-procedure learning game rather than a government-performance campaign simulator. It should enter later only if it exports comparable access-to-justice, procedural fairness, litigation-cost, or rule-compliance metrics.

12 Working Conclusion

The strongest current conclusion is methodological and provisional. Methodologically, institutional portfolios should be ranked under multiple value profiles, checked for interval overlap, filtered by exact, epsilon, and uncertainty-aware Pareto rules, tested through feedback loops, rerun under explicit stress profiles, and then screened through adaptive-survival gates before design claims are made. Substantively, the current configured runs make single-rule fixes look weak. The balanced winner is the leading static point-estimate and lower-bound candidate, but it is not uncertainty-resolved and it fails the adaptive recommendation gate. The robustness and adaptive gray-zone leaders are better validation targets than final designs. No portfolio should be recommended as settled policy until the adaptive assumptions are calibrated against external evidence.

A Technical Method Appendix

A.1 Static Portfolio Scoring

The cumulative score is built from normalized subsystem diagnostics rather than from one raw simulator score. The importer first aggregates each source report by scenario key. It then computes component summaries from available columns only; missing source columns are skipped rather than imputed. The portfolio layer combines one legislature, one review design, and one anti-capture design into diagnostics for policy delivery, public alignment, rights safeguard, capture resistance, legitimacy, efficiency, complexity score, and resilience floor.

The balanced score is intentionally not pure throughput. Delivery and efficiency matter, but the score also rewards rights safeguards, capture resistance, legitimacy, public alignment, and subsystem floor. The floor term is conservative: a portfolio that performs well in two subsystems but collapses on rights, capture resistance, or complexity should not become the headline candidate merely because its average is high.

Diagnostic	Formula summary
Policy delivery	Mean of legislature productivity, welfare, public alignment, inverse gridlock, and inverse time-to-correct-bad-law when available.
Public alignment	Mean of legislative representative quality, legislative public alignment, legislative and review democratic responsiveness, anti-capture representation, and anti-capture public interest.
Rights safeguard	Mean of review rights protection, legal stability, stability-rights score, inverse shadow-docket abuse, inverse emergency legitimacy risk, inverse constitutional conflict, inverse weak public-mandate passage, and inverse concentrated-harm passage.
Capture resistance	Mean of anti-capture control, anti-capture success, detection, sanctions, transparency gain, inverse capture rate, inverse public-preference distortion, inverse hidden influence, inverse legislative lobby capture, and legislative anti-lobbying success.
Legitimacy	Mean of legislative legitimacy, review legitimacy, review public confidence, review legitimacy-control score, and anti-capture representation.
Complexity score	Mean of administrative feasibility, inverse administrative costs, inverse review institutional cost, inverse implementation complexity, anti-capture reform feasibility, and inverse anti-capture administrative cost.
Efficiency	0.65 times policy delivery plus 0.35 times complexity score.
Resilience floor	Minimum of legislature component score, review component score, anti-capture component score, rights safeguard, capture resistance, and complexity score.

Table 15: Static portfolio diagnostic construction. Values are normalized to 0–1 before aggregation.

A.2 Profile Sensitivity

The alternate profiles expose normative dependence in the rankings. The efficiency-first profile rewards delivery, implementation efficiency, and low complexity. The rights-first profile rewards rights safeguards, legal stability, legitimacy, and resilience floor. The anti-capture-first profile emphasizes capture resistance, public alignment, legitimacy, and transparency. The low-complexity profile emphasizes administrative tractability. The legitimacy-first profile emphasizes legitimacy, democratic/public alignment, rights safeguards, and capture resistance. A portfolio that wins only one profile is treated as a value-specific candidate; a portfolio that remains competitive across profiles is treated as a stronger design candidate.

Profile	Weights
Balanced	policyDelivery 22%; publicAlignment 18%; rightsSafeguard 18%; captureResistance 18%; legitimacy 10%; efficiency 8%; resilienceFloor 6%
Efficiency First	policyDelivery 32%; efficiency 26%; complexityScore 18%; publicAlignment 8%; rightsSafeguard 6%; captureResistance 6%; legitimacy 4%
Rights First	rightsSafeguard 36%; legitimacy 18%; publicAlignment 14%; captureResistance 12%; policyDelivery 8%; resilienceFloor 8%; efficiency 4%
Anti-Capture First	captureResistance 38%; publicAlignment 16%; legitimacy 14%; rightsSafeguard 10%; policyDelivery 8%; resilienceFloor 8%; efficiency 6%
Low Complexity	complexityScore 34%; efficiency 20%; resilienceFloor 14%; policyDelivery 12%; publicAlignment 8%; rightsSafeguard 6%; captureResistance 6%
Legitimacy First	legitimacy 30%; publicAlignment 24%; rightsSafeguard 14%; captureResistance 12%; policyDelivery 8%; complexityScore 6%; resilienceFloor 6%

Table 16: Scoring profile weights used for static reranking.

Profile	Winning portfolio	Score	Balanced rank
Balanced	Expanded portfolio hybrid legislature + No emergency relief without merits review + Full anti-capture bundle	0.656	1
Efficiency First	Default pass unless 2/3 block + No emergency relief without merits review + Democracy vouchers	0.727	348
Rights First	Unicameral majority + pairwise alternatives + No emergency relief without merits review + Full anti-capture bundle	0.665	125
Anti-Capture First	Unicameral majority + pairwise alternatives + No emergency relief without merits review + Full anti-capture bundle	0.635	125
Low Complexity	Default pass unless 2/3 block + 60 percent invalidation threshold + Democracy vouchers	0.723	645
Legitimacy First	Unicameral majority + pairwise alternatives + No emergency relief without merits review + Full anti-capture bundle	0.627	125

Table 17: Profile-specific portfolio winners.

Focused case	B	Eff.	Rights	Cap.	Low-cx	Legit.	Regret
balanced winner	1	2,245	44	3	25,216	34	5,956
Rights First winner	125	4,183	1	1	5,487	1	1,700
robustness winner	267	4,727	14	10	4,208	7	2,285
minimax-regret winner	323	1,122	3,877	117	1,318	672	1
Efficiency First winner	348	1	9,334	578	5	1,439	39
Low Complexity winner	645	3	17,601	1,155	1	6,205	418
current-system-ish baseline	33,486	30,635	33,054	31,561	17,043	33,439	29,727

Table 18: Profile-by-profile ranks for the focused candidate set. Lower ranks are better.

Case	Balanced rank	Score	Synthetic band	Half-width	Max regret
balanced winner	1	0.656	0.563–0.748	0.092	0.107
Rights First winner	125	0.640	0.548–0.733	0.092	0.083
robustness winner	267	0.635	0.542–0.727	0.092	0.086
minimax-regret winner	323	0.633	0.538–0.727	0.094	0.051
Efficiency First winner	348	0.631	0.533–0.730	0.098	0.061
Low Complexity winner	645	0.625	0.526–0.723	0.099	0.073
current-system-ish baseline	33,486	0.500	0.396–0.603	0.104	0.163

Table 19: Uncertainty and regret diagnostics for the focused candidate set. Bands are synthetic model-sensitivity bands, not empirical confidence intervals.

A.3 Robustness and Pareto Filtering

The robustness report is separate from the balanced report. Its score rewards low average profile rank, low worst profile rank, repeated top-profile appearances, limited score spread, and resilience floor. This is why the robustness winner can rank lower under the balanced score but remain central to the paper: it tests whether a design survives reasonable changes in value weighting.

The Pareto filter identifies portfolios that are not clearly dominated across the major diagnostics. It does not claim that every Pareto-front portfolio is equally plausible. Its role is narrower: it prevents the paper from discarding designs that are weaker on the headline score but stronger on a constitutionally important dimension such as rights, legitimacy, capture resistance, or complexity.

Regret rank	Max regret	Avg. regret	Balanced rank	Portfolio
1	0.051	0.034	323	Default pass + multi-round mediation + challenge + No emergency relief without merits review + Full anti-capture bundle
2	0.053	0.036	354	Default pass + multi-round mediation + challenge + Automatic merits follow-up for emergency relief + Full anti-capture bundle
3	0.054	0.023	119	Default pass unless 2/3 block + No emergency relief without merits review + Full anti-capture bundle
4	0.054	0.024	127	Default pass unless 2/3 block + Emergency integrity package + Full anti-capture bundle
5	0.055	0.035	327	Default pass + multi-round mediation + challenge + Emergency integrity package + Full anti-capture bundle
6	0.056	0.038	446	Default pass + multi-round mediation + challenge + Randomized merits panels with en banc correction + Full anti-capture bundle
7	0.057	0.038	447	Default pass + multi-round mediation + challenge + Three-judge panels with en banc correction + Full anti-capture bundle
8	0.057	0.039	464	Default pass + multi-round mediation + challenge + Comparative 16-seat constitutional senates + Full anti-capture bundle

Table 20: Minimax-regret leaders across the configured value profiles. Low regret does not by itself make a portfolio recommendable.

Robustness diagnostic	Value
Portfolio	Portfolio hybrid legislature + No emergency relief without merits review + Full anti-capture bundle
Robust score	0.761
Balanced rank	267
Average profile rank	1538.833
Worst profile rank	4,727
Top-25 profile appearances	3

Table 21: Cross-profile robustness winner.

Balanced rank	Regret rank	Band	Portfolio	Do-not-recommend-yet reason
1	5,956	0.092	Expanded portfolio hybrid legislature + No emergency relief without merits review + Full anti-capture bundle	weak subsystem floor; weak public alignment; falls far under at least one value profile
2	6,433	0.092	Expanded portfolio hybrid legislature + Emergency integrity package + Full anti-capture bundle	weak subsystem floor; weak public alignment; falls far under at least one value profile
3	6,183	0.092	Expanded portfolio hybrid legislature + Automatic merits follow-up for emergency relief + Full anti-capture bundle	weak subsystem floor; weak public alignment; falls far under at least one value profile
4	7,130	0.092	Expanded portfolio hybrid legislature + Constitutional council with concrete-review backstop + Full anti-capture bundle	weak subsystem floor; weak public alignment; falls far under at least one value profile
5	6,870	0.092	Expanded portfolio hybrid legislature + Pre-enactment constitutional council + Full anti-capture bundle	weak subsystem floor; weak public alignment; falls far under at least one value profile
6	6,977	0.092	Expanded portfolio hybrid legislature + Constitutional remand before invalidation + Full anti-capture bundle	weak subsystem floor; weak public alignment; falls far under at least one value profile
7	7,722	0.092	Expanded portfolio hybrid legislature + Constitutional remand with override window + Full anti-capture bundle	weak subsystem floor; weak public alignment; falls far under at least one value profile
8	6,047	0.092	Expanded portfolio hybrid legislature + Three-judge panels with en banc correction + Full anti-capture bundle	weak subsystem floor; weak public alignment; falls far under at least one value profile

Table 22: Fragile headline or near-headline portfolios that should not be converted into recommendations without more modeling.

Pareto diagnostic	Value
Total generated portfolios	33,852
Exact point-estimate Pareto front	2,290 (6.8%)
Epsilon-Pareto front	203 (0.6%)
Uncertainty-aware Pareto front	16,673 (49.3%)

Table 23: Pareto-front sizes under exact, epsilon, and uncertainty-aware dominance rules.

A.4 Bridge Coefficients and Stress Profiles

The bridge model is deterministic and configured through `config/interbranch-bridge-v0.json`. Its coefficients link legislative delivery, bad-policy inflow, review docket pressure, correction, court-curbing pressure, capture drift, public-alignment erosion, legitimacy, and compliance. The bridge does not rerun the sibling simulators period by period; it uses the portfolio diagnostics as initial conditions and feedback parameters.

The stress profiles are scenario multipliers over that same bridge. High-capture pressure raises starting capture and capture growth. Low-public-trust weakens starting legitimacy and compliance. High-rights-threat stress increases review pressure and quality plus backlog penalties. Court-curbing and noncompliance stress increases retaliation and compliance decay. Administrative-complexity stress reduces effective delivery, review capacity, and compliance. These stresses are not empirical forecasts; they are structured sensitivity checks over the same focused comparison set.

Baseline bridge rank	Focused case	Static balanced rank	Selector
1	Balanced winner	1	cumulative-government-portfolios.csv rank 1
2	Efficiency caution case	348	profile-sensitivity.csv efficiency-first rank 1
3	Minimax-regret winner	323	portfolio-minimax-regret.csv rank 1
4	Rights/capture winner	125	profile-sensitivity.csv rights-first rank 1
5	Legitimacy-first winner	125	profile-sensitivity.csv legitimacy-first rank 1
6	Robustness winner	267	portfolio-robustness.csv rank 1
7	Current-system-ish baseline	33,486	fixed baseline keys

Table 24: Bridge v0 evaluates 7 focused cases from 33,852 static portfolios; 33,845 static portfolios are not bridge-simulated in the paper workflow.

Win/top-two rank	Case	Avg. rank	Worst	Wins	Top-two	Avg. score	Max regret
1	Balanced winner	2.250	5	5	5	0.593	0.289
2	Robustness winner	4.125	6	2	2	0.661	0.015
3	Rights/capture winner	2.625	4	1	5	0.660	0.033
4	Efficiency caution case	4.625	7	0	3	0.556	0.314
5	Legitimacy-first winner	3.625	5	0	1	0.660	0.033
6	Minimax-regret winner	4.125	6	0	0	0.570	0.298
7	Current-system-ish baseline	6.625	7	0	0	0.379	0.483

Table 25: Stress-profile stability across the focused bridge cases, sorted by the wins/top-two rule.

A.5 Adaptive Bridge v1

Adaptive Bridge v1 is configured through `config/adaptive-bridge-v1.json`. It reuses the v0 stress profiles but expands the state vector to include transition load, sequencing readiness, strategic adaptation by parties, courts, and lobbyists, federalism resistance, delivery bottlenecks, recovery capacity, agency implementation gaps, voter feedback pressure, feedback correction capacity, emergency safeguards, and citizen agenda capacity. Its recommendation gate requires a strong average adaptive score, limited worst-stress rank, limited adaptive regret, bounded synthetic uncertainty,

Criterion	Winner	Winner metric
Wins/top-two rule	Balanced winner	5 wins; 5 top-two placements; average rank 2.250
Best average stress rank	Balanced winner	average rank 2.250; worst rank 5
Best average bridge score	Robustness winner	average bridge score 0.661
Lowest maximum stress regret	Robustness winner	max stress regret 0.015; average regret 0.009
Smallest score spread	Robustness winner	score spread 0.236

Table 26: Alternative bridge-stability definitions. The wins/top-two rule is the default report sort, not the only stability interpretation.

Bridge-score component	Weight
policyQuality	22%
legitimacy	18%
captureControl	17%
backlogControl	14%
compliance	12%
averageDelivery	10%
courtCurbingControl	7%

Table 27: Bridge-score weights from `config/interbranch-bridge-v0.json`.

Stress profile	Modifiers
Baseline	none
High Capture Pressure	initialCapturePressureAdd=0.16; captureGrowthMultiplier=1.35; captureControlMultiplier=0.78; publicAlignmentCapturePenaltyMultiplier=1.25; legitimacyCapturePenaltyMultiplier=1.2; captureQualityPenaltyMultiplier=1.25
Low Public Trust	baseLegitimacyMultiplier=0.82; initialComplianceAdd=-0.08; complianceMultiplier=0.9; legitimacyUncorrectedPenaltyMultiplier=1.25; legitimacyCapturePenaltyMultiplier=1.15; deliveryMultiplier=0.96
High Rights-Threat Environment	badPolicyMultiplier=1.2; rightsThreatMultiplier=1.45; docketPressureMultiplier=1.2; badPolicyBacklogMultiplier=1.2; badPolicyQualityPenaltyMultiplier=1.25; netBenefitBadPolicyPenaltyMultiplier=1.2
Court-Curbing and Institutional Noncompliance Stress	initialCourtCurbingPressureAdd=0.1; courtCurbingGrowthMultiplier=1.45; courtCurbingDecayMultiplier=0.65; complianceMultiplier=0.86; reviewCurbingPenaltyMultiplier=1.3; legitimacyCurbingPenaltyMultiplier=1.35; deliveryCourtCurbingPenaltyMultiplier=1.25
Emergency Abuse Stress	initialCorrectionBacklogAdd=0.08; badPolicyMultiplier=1.16; rightsThreatMultiplier=1.38; docketPressureMultiplier=1.28; reviewCapacityMultiplier=0.92; badPolicyBacklogMultiplier=1.18; uncorrectedBacklogMultiplier=1.18; badPolicyQualityPenaltyMultiplier=1.2; legitimacyUncorrectedPenaltyMultiplier=1.32; netBenefitUncorrectedPenaltyMultiplier=1.2
Administrative Overload and Complexity Stress	baseComplexityMultiplier=0.82; baseFloorMultiplier=0.92; deliveryMultiplier=0.92; reviewCapacityMultiplier=0.88; complianceMultiplier=0.92; captureGrowthMultiplier=1.08; docketPressureMultiplier=1.08
Federalism and Agency-Capacity Stress	baseComplexityMultiplier=0.78; baseFloorMultiplier=0.9; deliveryMultiplier=0.88; reviewCapacityMultiplier=0.9; complianceMultiplier=0.84; captureGrowthMultiplier=1.12; docketPressureMultiplier=1.12; publicAlignmentUncorrectedPenaltyMultiplier=1.18; legitimacyUncorrectedPenaltyMultiplier=1.18; deliveryCourtCurbingPenaltyMultiplier=1.15

Table 28: Bridge stress modifiers. Multipliers and additive adjustments are applied to the baseline bridge equations.

bounded profile regret, an adequate resilience floor, and enough direct evidence not to treat a thin family-specific result as recommendable.

The gate has four meanings. A provisional shortlist row passes the current synthetic screen but is still not an empirical recommendation. A calibration gray-zone row is strong enough to deserve direct validation but has unresolved evidence, stress, regret, or failure-mode caveats. A review-priority row is worth further modeling but fails more of the current gate conditions. A do-not-recommend-yet row should be treated as fragile under the present assumptions.

Synthetic adaptive mechanism introduced in Bridge v1
transition load from reform complexity, low subsystem floor, uncertainty, and profile regret
sequencing readiness from front-loaded legality, staffing, transparency, and review capacity
party adaptation and agenda obstruction
court adaptation and court-curbing pressure
lobby adaptation and venue shifting across agencies, courts, procurement, ballot campaigns, think tanks, and associations
agency implementation gaps under federalism resistance, delivery bottlenecks, and recovery capacity
public feedback through asymmetric trust erosion, voter backlash, compliance, and legitimacy
citizen agenda capacity as a corrective channel, with overload and rights-risk penalties for weakly deliberative tools
emergency-powers safeguards against fast, evasive, or normalized executive action
family-level direct-evidence strength used as a recommendation gate caveat

Table 29: Adaptive Bridge v1 mechanisms. These are transparent synthetic equations, not empirically calibrated parameters.

Calibration area	Use in Adaptive Bridge v1
Lobby venue shifting	Anti-capture rows with audit, machine-readable logs, venue-shift detection, or sanctions reduce lobby adaptation and capture pressure; narrow caps and budget tools keep a substitution-risk caveat.
Emergency powers	Emergency-integrity rows reduce rights-threat inflow, court adaptation, court-curbing pressure, and review backlog only when review is fast, reason-giving, and anti-evasion oriented.
Sequencing and transition costs	Sequencing readiness reduces transition load, agency gaps, and delivery bottlenecks; complex one-shot redesigns carry higher transition and direct-evidence penalties.
Federalism and agency capacity	The bridge now tracks federalism resistance, delivery bottleneck pressure, and recovery capacity separately before combining them into agency implementation gaps.
Public feedback	Feedback correction capacity can turn backlash into alignment and policy correction, while weak voice causes faster legitimacy erosion.
Citizen agenda setting	Citizen assemblies, random public review panels, and comment-authenticity rules improve agenda correction; initiatives and petition channels also add rights-risk or overload pressure.
Comparative evidence strength	Low-direct-evidence families can rank highly synthetically, but the recommendation gate labels them as calibration gray-zone or do-not-recommend rather than provisional recommendations.

Table 30: Evidence-informed priors added to Adaptive Bridge v1. The table records directional model use only; it is not an empirical fit.

A.6 Input Provenance

The paper build records hashes for the imported sibling simulator artifacts. The table below gives abbreviated hashes for print readability; the full SHA-256 values are written beside the generated LaTeX fragments.

Coefficient family	Evidence tier	Additive coefficients	Multiplier coefficients
audit-and-sanctions anti-capture	evidence-anchored prior	capCtrl=0.060; recovery=0.020; transReady=0.025	lobbyAdapt=0.920
venue-shifting detection	evidence-anchored prior	capCtrl=0.050; feedback=0.020; transReady=0.035	lobbyAdapt=0.880
democracy vouchers	evidence-informed prior	agendaCap=0.040; capCtrl=0.020; feedback=0.025	lobbyAdapt=0.980
budgeted disclosed lobbying	low-direct-evidence prior	capCtrl=0.025; capPress=0.012; transReady=0.010	lobbyAdapt=0.980
intermediary and dark-money substitution	evidence-anchored risk prior	agendaLoad=0.020; capPress=0.050	lobbyAdapt=1.120
emergency integrity	evidence-anchored prior	courtCurb=0.045; emergSafe=0.100; recovery=0.030; reviewCap=0.050; transReady=0.040	–
jurisdiction and pre-enactment review	evidence-informed prior	courtCurb=0.030; emergSafe=0.040; reviewCap=0.030; transReady=0.030	–
citizen assembly threshold	evidence-informed prior	agendaCap=0.070; agendaLoad=0.025; feedback=0.055; transReady=0.020	partyAdapt=0.960
random public review panel	evidence-informed prior	agendaCap=0.060; feedback=0.050; rightsRisk=-0.025	partyAdapt=0.970
citizens agenda petition	speculative-to-evidence-informed prior	agendaCap=0.060; agendaLoad=0.035; feedback=0.035	partyAdapt=0.970
citizen initiative referendum	risk-weighted prior	agendaCap=0.080; agendaLoad=0.050; capPress=0.020; feedback=0.025; rightsRisk=0.045	–
authenticated public participation	evidence-informed prior	agendaCap=0.035; capCtrl=0.030; feedback=0.035; transReady=0.020	–
pairwise alternatives	low-direct-evidence prior	deliverGap=0.015; feedback=0.025; transLoad=0.035	partyAdapt=0.980
portfolio hybrid	low-direct-evidence prior	deliverGap=0.020; transLoad=0.045; transReady=-0.020	–
default pass	risk-weighted prior	capPress=0.020; partyAdapt=0.020; rightsRisk=0.060	–

Table 31: Configured Adaptive Bridge v1 coefficient families. Coefficients are loaded from the v1 configuration and are evidence-anchored priors, not fitted causal estimates.

Adaptive rank	Avg. score	Worst stress rank	Max regret	Evidence	Gate	Portfolio
1	0.705	34	0.021	mixed; low-direct-evidence family included	calibration gray zone, not recommendation	Unicameral majority + pairwise alternatives + No emergency relief without merits review + Full anti-capture bundle
2	0.704	37	0.022	mixed; low-direct-evidence family included	calibration gray zone, not recommendation	Pairwise amendment tournament + majority + No emergency relief without merits review + Full anti-capture bundle
3	0.703	66	0.027	mixed; low-direct-evidence family included	calibration gray zone, not recommendation	Unicameral majority + pairwise alternatives + Emergency integrity package + Full anti-capture bundle
4	0.703	70	0.028	mixed; low-direct-evidence family included	calibration gray zone, not recommendation	Pairwise amendment tournament + majority + Emergency integrity package + Full anti-capture bundle
5	0.701	109	0.032	low direct evidence	calibration gray zone, not recommendation	Unicameral majority + pairwise alternatives + Automatic merits follow-up for emergency relief + Full anti-capture bundle
6	0.700	116	0.033	low direct evidence	calibration gray zone, not recommendation	Pairwise amendment tournament + majority + Automatic merits follow-up for emergency relief + Full anti-capture bundle
7	0.700	117	0.033	mixed; low-direct-evidence family included	calibration gray zone, not recommendation	Portfolio hybrid legislature + No emergency relief without merits review + Full anti-capture bundle
8	0.698	346	0.020	evidence-informed	calibration gray zone, not recommendation	Citizen initiative and referendum + No emergency relief without merits review + Full anti-capture bundle
9	0.698	184	0.039	mixed; low-direct-evidence family included	calibration gray zone, not recommendation	Portfolio hybrid legislature + Emergency integrity package + Full anti-capture bundle
10	0.698	249	0.044	low direct evidence	calibration gray zone, not recommendation	Unicameral majority + pairwise alternatives + Pre-enactment constitutional council + Full anti-capture bundle

Table 32: Adaptive Bridge v1 all-portfolio leaders. Gate status is a synthetic screening result, not a recommendation.

Stress profile	Adaptive winner	Score	Static rank
Administrative Overload and Complexity Stress	Default pass unless 2/3 block + No emergency relief without merits review + Full anti-capture bundle	0.687	119
Baseline	Default pass unless 2/3 block + No emergency relief without merits review + Full anti-capture bundle	0.742	119
Court-Curbing and Institutional Noncompliance Stress	Default pass unless 2/3 block + No emergency relief without merits review + Full anti-capture bundle	0.721	119
Emergency Abuse Stress	Unicameral majority + pairwise alternatives + No emergency relief without merits review + Full anti-capture bundle	0.739	125
Federalism and Agency-Capacity Stress	Citizen initiative and referendum + No emergency relief without merits review + Full anti-capture bundle	0.646	1,420
High Capture Pressure	Unicameral majority + pairwise alternatives + No emergency relief without merits review + Full anti-capture bundle	0.714	125
High Rights-Threat Environment	Unicameral majority + pairwise alternatives + No emergency relief without merits review + Full anti-capture bundle	0.740	125
Low Public Trust	Default pass unless 2/3 block + No emergency relief without merits review + Full anti-capture bundle	0.692	119

Table 33: Stress-profile winners in Adaptive Bridge v1 over all static portfolios.

Source	Rows	Scenarios	Imported report	SHA-256 prefix
Congress Institutional Simulator	1,428	42	`simulation-campaign-v21-paper.csv`	`1e36005c54d3a0cc`
Supreme Court Simulator Design	520	26	`constitutional-review-campaign-v2.csv`	`2278a169405c38ec`
Lobby Capture Simulator	31	31	`lobby-capture-campaign.csv`	`8e1e313e166cb8a6`

Table 34: Input source provenance for the current paper build. Full hashes are written to `paper/generated/source-provenance.json`.